Translation Constraint Joint
with
Five Degrees of Freedom

A Report

by

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CAD MODEL

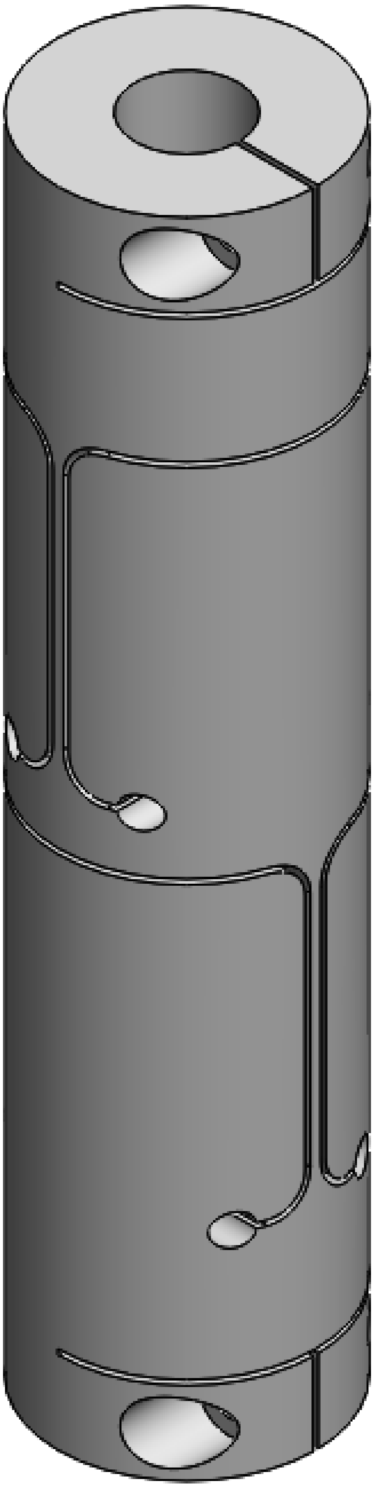


Figure 1: Isometric View

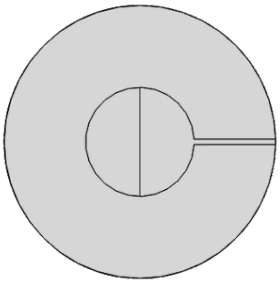


Figure 2: Top View

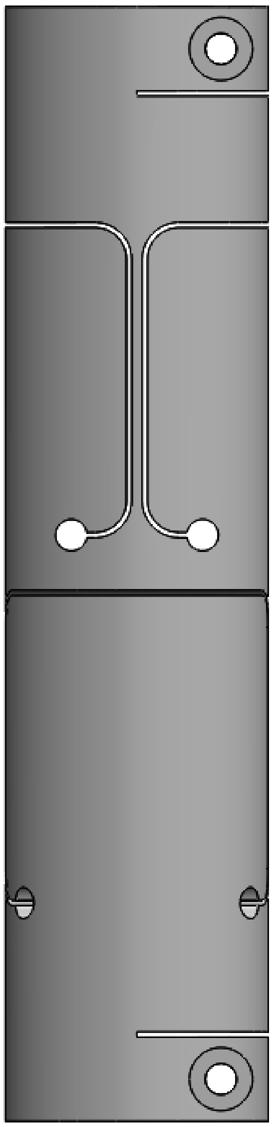


Figure 3: Front View

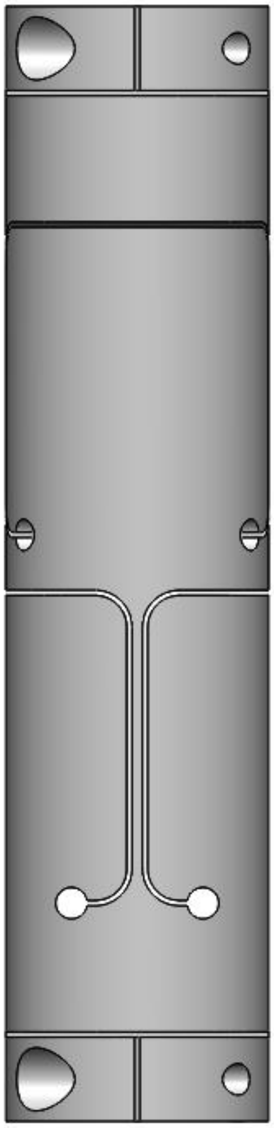
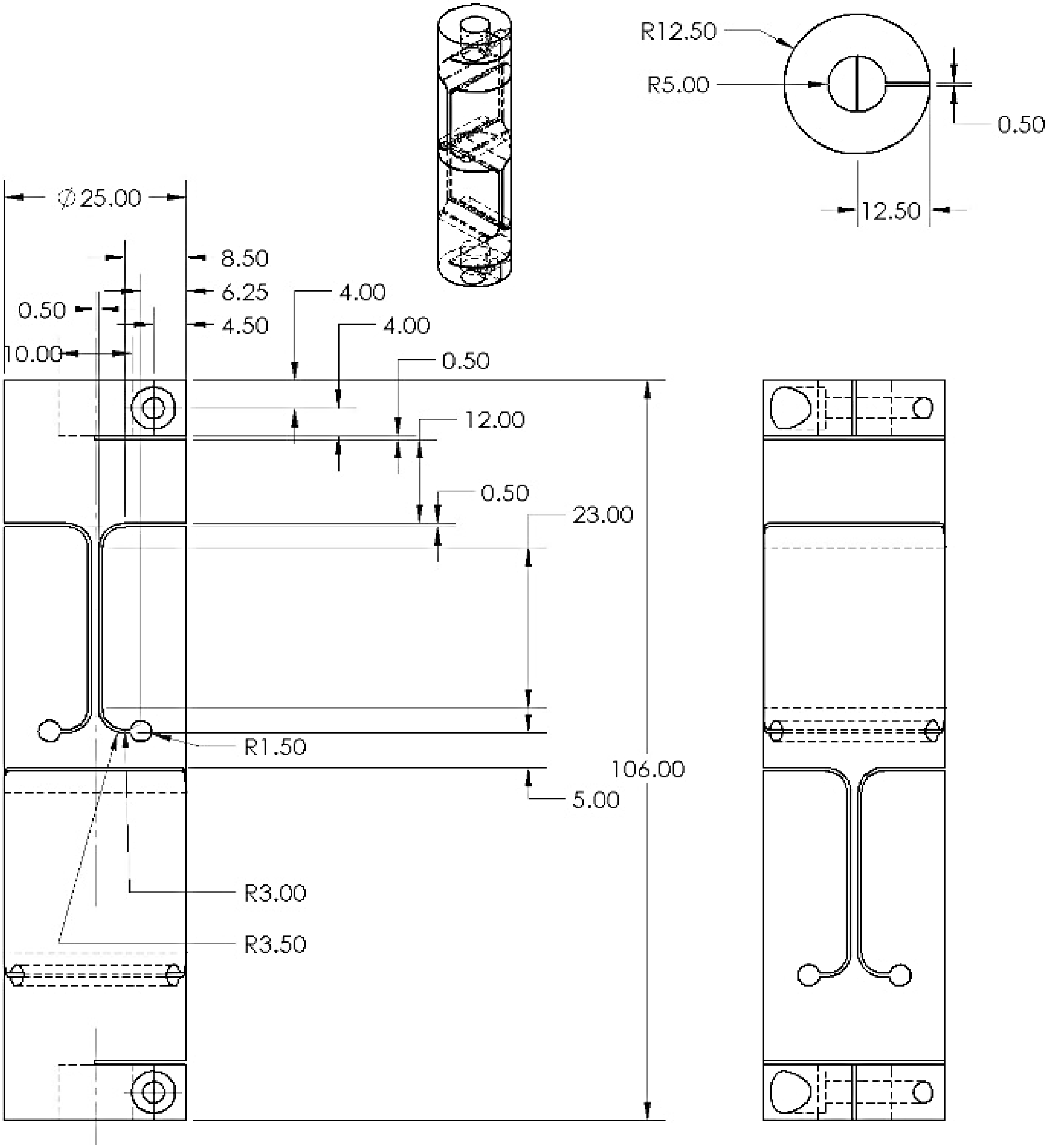


Figure 4: Right View

CAD Drawing



R12.50
R5.00
0.50
12.50
Ø25.00
8.50
6.25
4.50
0.50
10.00
4.00
4.00
0.50
12.00
0.50
23.00
R1.50
106.00
5.00
R3.00
R3.50

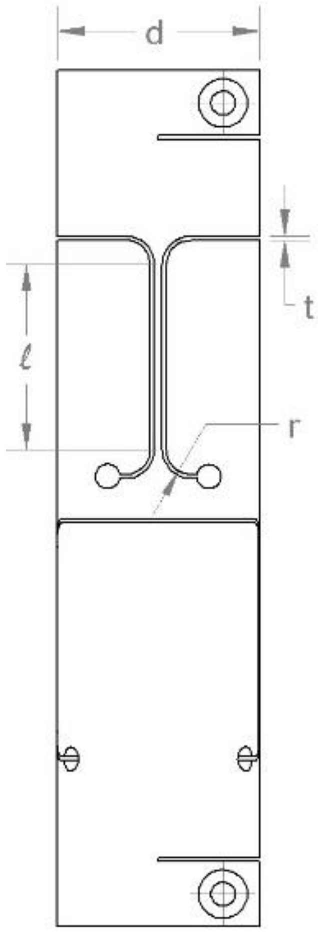
Specifications

Motion	Displacement Range	Stiffness		Maximum Stress
		Simulation	Analytical	
Rotation in X or Z Axis	±2.2º	16.6 Nm/rad	18.1 Nm/rad	167.36 MPa
Rotation in Y Axis	±4.6º	21.86 Nm/rad	21.81 Nm/rad	308.35 MPa
Translation in X Axis	±0.5 mm	35.3 N/mm	40.6 N/mm	186.61 MPa
Translation in Z Axis	±0.5 mm	19.318 N/mm	21.34 N/mm	162.13 MPa
Translation in Y Axis	Constraint	84745.76 N/mm	94339.62 N/mm	-

Range of motion with changing parameters

1. Rotation in X or Z Axis:

$$\theta = \sin^{-1}\left(\frac{l+r+t}{\sqrt[2]{(l+r)^2+\left(\frac{d}{2}\right)^2}}\right) - \sin^{-1}\left(\frac{l+r}{\sqrt[2]{(l+r)^2+\left(\frac{d}{2}\right)^2}}\right)$$



2. Rotation in Y Axis:

$$\theta = \sin^{-1}\left(\frac{2t}{\sqrt[2]{(t)^2+\left(\frac{d}{2}\right)^2}}\right) - \sin^{-1}\left(\frac{t}{\sqrt[2]{(t)^2+\left(\frac{d}{2}\right)^2}}\right)$$

3. Translation in X or Z Axis:

$$\delta = \pm t$$

1. Rotation in X or Z Axis

Torque = 665 Nmm

θ = 0.0399 rad (from simulation)

Using the formula, $\theta = \frac{ML}{EI}$, for slope due to a moment, we get the value θ = 0.0367 rad

Value of stiffness calculated from simulation, K= 16.6 Nm/rad

Value of stiffness calculated from formula, K= 18.1 Nm/rad

Percentage error = 9%

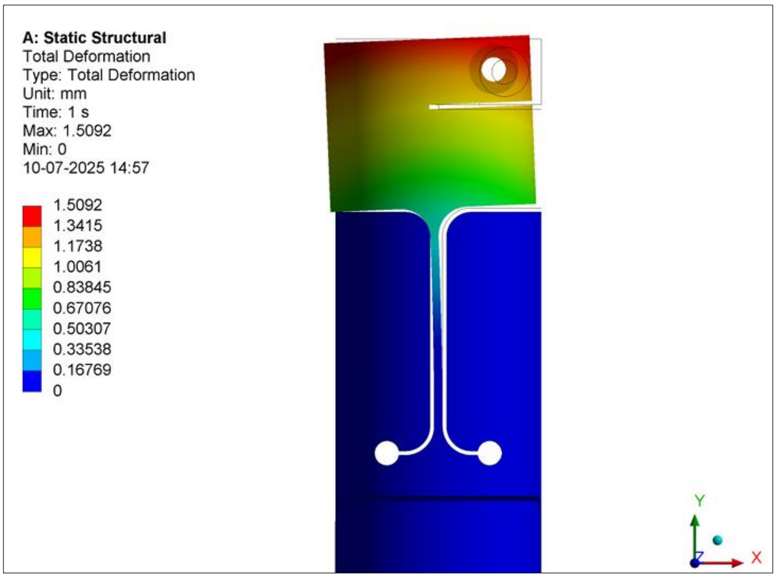


Figure 5: Displacement

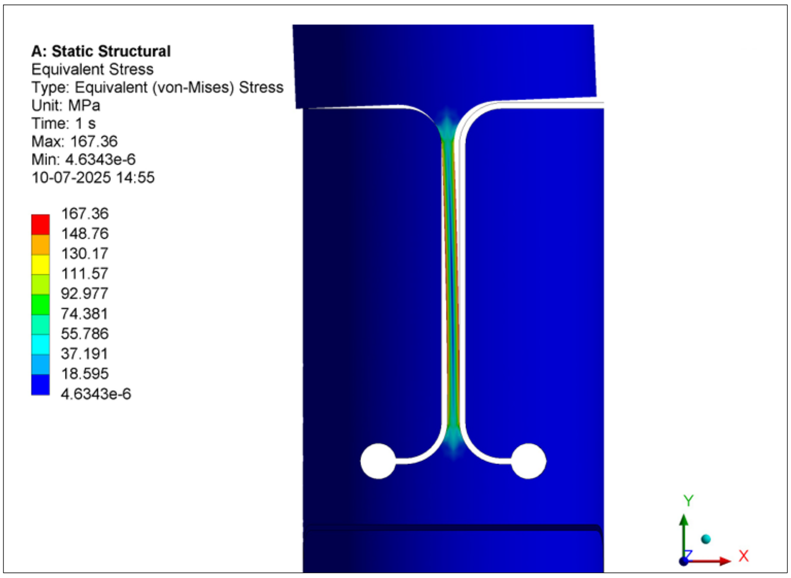


Figure 6: Stress

2. Rotation in Y Axis

Torque = 1800 Nmm

θ = 0.0823 rad (from simulation)

Using the formula, $k = ab^3\left[\frac{16}{3} - 3.36\frac{b}{a}\left(1 - \frac{b^4}{12a^4}\right)\right] \ldots (a \geq b)$, for coefficient of stiffness, we get k = 129.973

Also, putting the value of k in the formula, $k = \frac{Tl}{\theta G}$, we get the value of stiffness i.e. $\frac{T}{\theta} = \frac{kG}{l}$, but this value of stiffness would be just for a single section, and in order to find the net stiffness we need to multiply it by 2.

Value of stiffness calculated from simulation, K= 21.86 Nm/rad

Value of stiffness calculated from formula, K= 21.81 Nm/rad

Percentage error = 0.2%

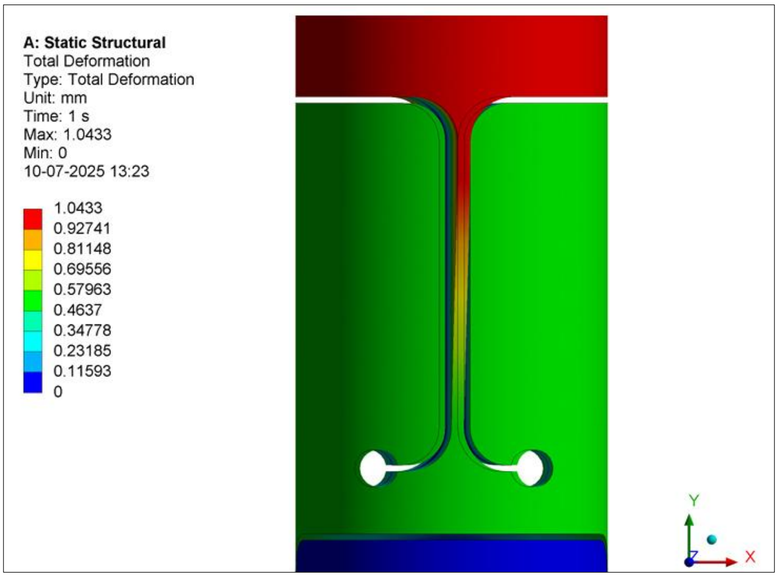


Figure 7: Displacement

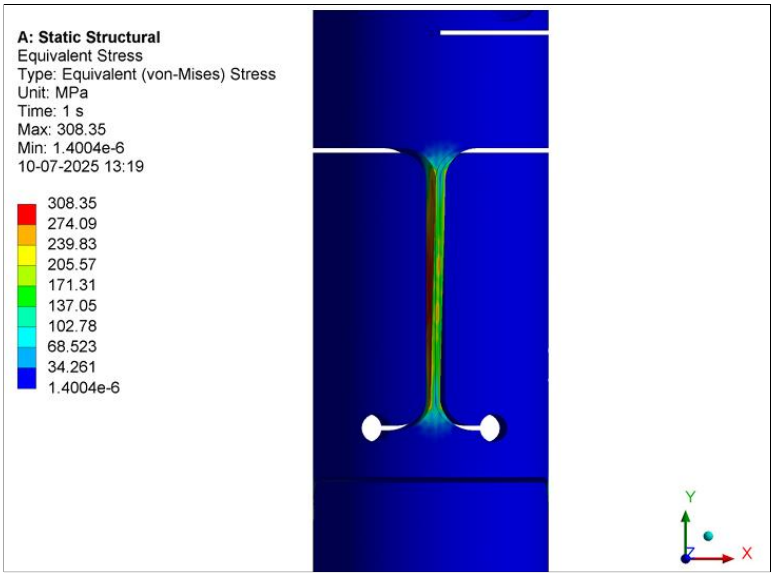


Figure 8: Stress

3. Translation in X Axis

Force = 17 N

δ = 0.4805 mm (from simulation)

Using the formula, $\delta = \frac{ML^2}{2EI}$, for deflection due to a moment, we get the value δ = 0.253 mm

And using the formula, $\delta = \frac{PL^3}{3EI}$, for deflection due to a load, we get the value δ = 0.165 mm

$\delta_{net} = 0.418$ mm

Value of stiffness calculated from simulation, K= 35.3 N/mm

Value of stiffness calculated from formula, K= 40.6 N/mm

Percentage error = 15%

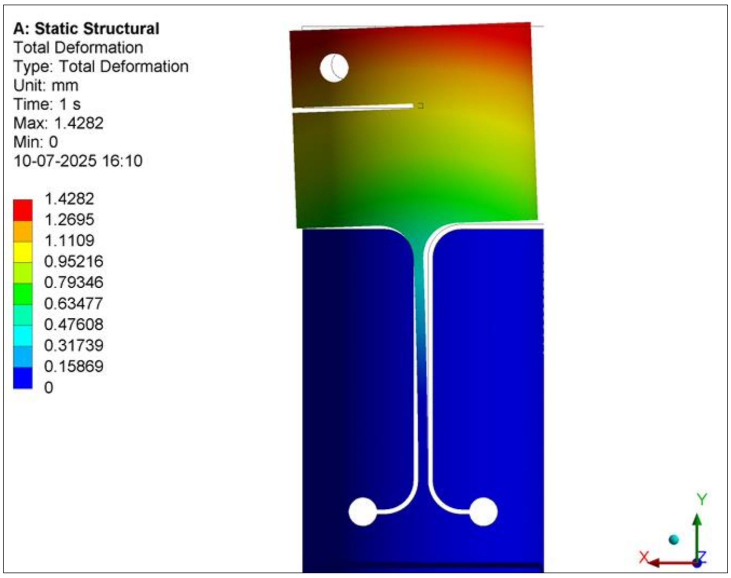


Figure 9: Displacement

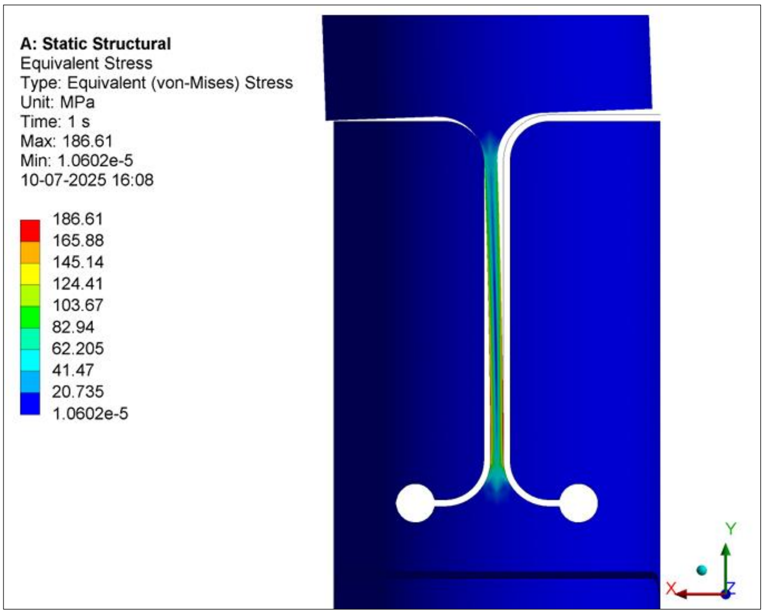


Figure 10: Stress

4. Translation in Z Axis

Force = 8.5 N

δ = 0.440 mm (from simulation)

Using the formula, $\delta = \frac{ML^2}{2EI}$, for deflection due to a moment, we get the value δ = 0.315 mm

And using the formula, $\delta = \frac{PL^3}{3EI}$, for deflection due to a load, we get the value δ = 0.0827 mm

δ_{net} = 0.3983 mm

Value of stiffness calculated from simulation, K= 19.318 N/mm

Value of stiffness calculated from formula, K= 21.34 N/mm

Percentage error = 10%

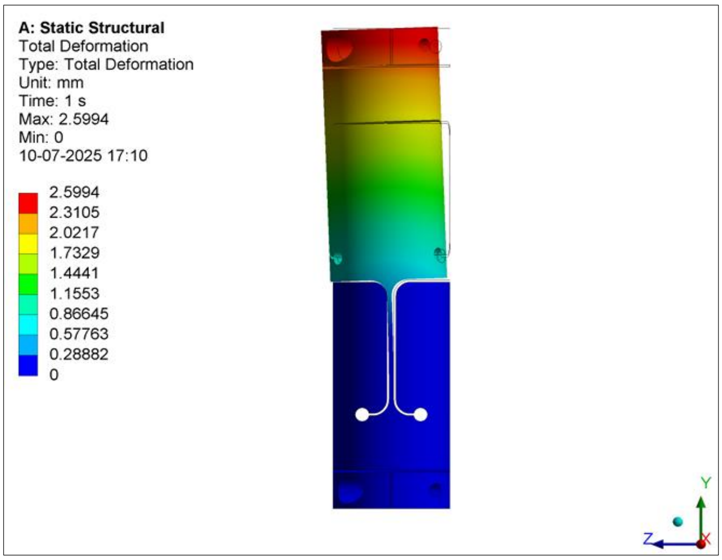


Figure 11: Displacement

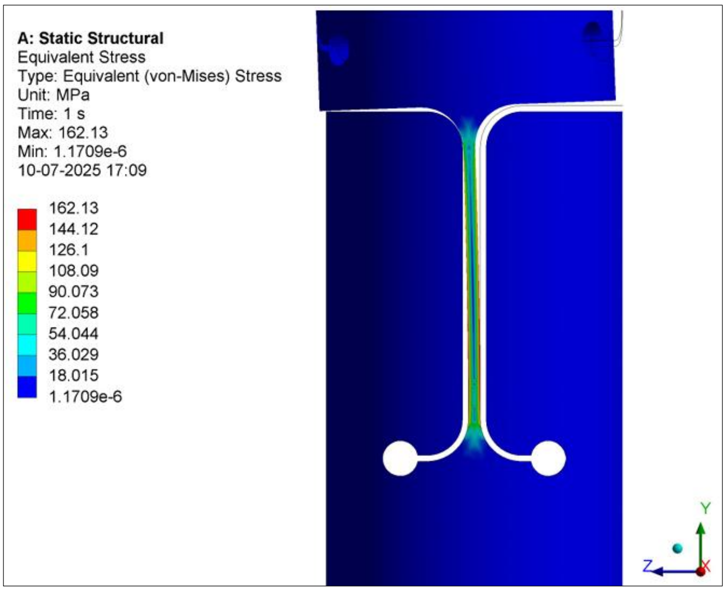


Figure 12: Stress

5. Translation in Y Axis

Force = 50 N

δ = 5.9 x 10^{-4} mm (from simulation)

Using the formula, $\delta = \frac{PL}{AE}$, we get the value δ = 5.3 x 10^{-4} mm

Value of stiffness calculated from simulation, K= 84745.76 N/mm

Value of stiffness calculated from formula, K= 94339.62 N/mm

Percentage error = 11.3%

Using the Euler's formula, $P = \frac{\pi^2 EI}{4L^2}$, for a column fixed at one end and free at the other, we can calculate the value of the

Critical Buckling Load = 2.3 x 10^4 N

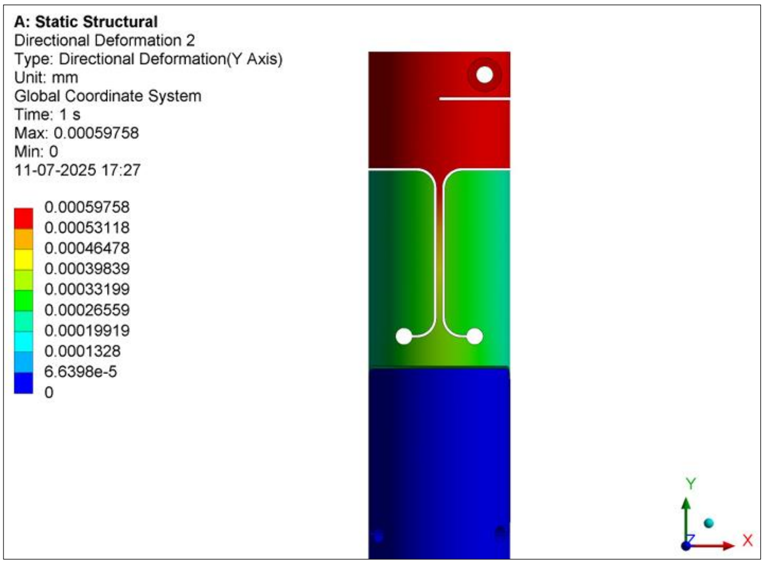


Figure 13: Displacement

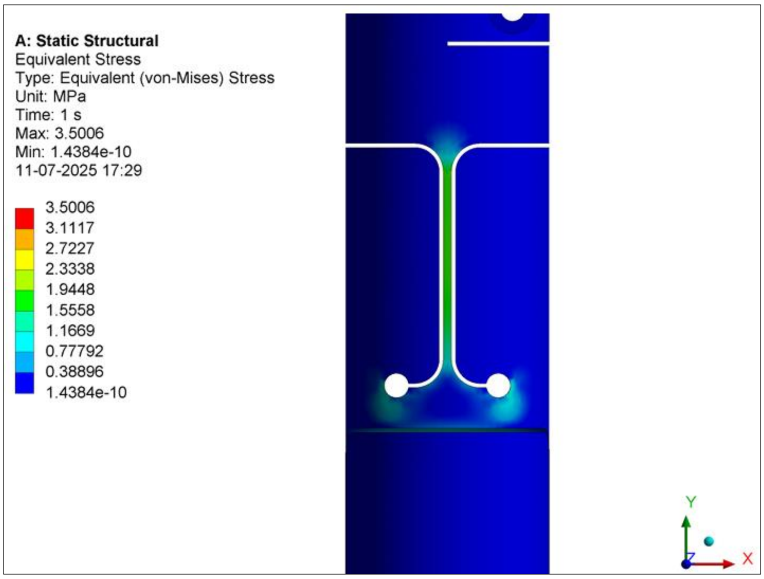


Figure 14: Stress